

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  struct Stock {
6      string symbol;
7      double price;
8      double change; // Price change from previous price
9  };
10
11 // Insertion Sort by change descending
12 void insertionSort(Stock arr[], int n) {
13     for (int i = 1; i < n; i++) {
14         Stock key = arr[i];
15         int j = i - 1;
16
17         // Move elements with smaller change to the right
18         while (j >= 0 && arr[j].change < key.change) {
19             arr[j + 1] = arr[j];
20             j--;
21         }
22         arr[j + 1] = key;
23     }
24 }
25
26 // Function to update stock price and calculate change
27 void updateStockPrice(Stock &stock, double newPrice) {
28     stock.change = newPrice - stock.price;
29     stock.price = newPrice;
30 }
31
32 // Print the list of stocks
33 void printStocks(Stock arr[], int n) {
```

EXPLORER

DSA

> .vscode

array.cpp

array.exe

event_scheduler.cpp

event_scheduler.exe

linkedlists.cpp

linkedlists.exe

sorting.cpp

sorting.exe

stock_ticker.cpp

stock_ticker.exe

cpp_soumya.cpp

sorting.cpp

stock_ticker.cpp

event_scheduler.cpp

sorting.exe

linkedlists.cpp

array.cpp

all.cpp

stock_ticker.cpp

Stock

```
31
32 // Print the list of stocks
33 void printStocks(Stock arr[], int n) {
34     cout << "Symbol\tPrice\tChange\n";
35     cout << "-----\n";
36     for (int i = 0; i < n; i++) {
37         cout << arr[i].symbol << "\t" << arr[i].price << "\t"
38             << (arr[i].change >= 0 ? "+" : "") << arr[i].change << "\n";
39     }
40     cout << "\n";
41 }
42
43 int main() {
44     const int n = 5;
45     Stock stocks[n] = {
46         {"AAPL", 178.23, 0},
47         {"GOOGL", 134.56, 0},
48         {"TSLA", 623.50, 0},
49         {"NFLX", 290.10, 0},
50         {"AMZN", 3450.00, 0}
51     };
52
53     cout << "Initial stock list:\n";
54     printStocks(stocks, n);
55
56     // Simulate some price updates
57     updateStockPrice(stocks[1], 135.00); // GOOGL +0.44
58     updateStockPrice(stocks[2], 620.00); // TSLA -3.50
59     updateStockPrice(stocks[4], 3460.00); // AMZN +10.00
60
61     // Sort stocks by change after updates
62     insertionSort(stocks, n);
63 }
```

I

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

- DSA
 - .vscode
 - array.cpp
 - array.exe
 - event_scheduler.cpp
 - event_scheduler.exe
 - linkedlists.cpp
 - linkedlists.exe
 - sorting.cpp
 - sorting.exe
 - stock_ticker.cpp
 - stock_ticker.exe

```
stock_ticker.cpp > main()
43 int main() {
58     updateStockPrice(stocks[2], 620.00); // TSLA -3.50
59     updateStockPrice(stocks[4], 3460.00); // AMZN +10.00
60
61     // Sort stocks by change after updates
62     insertionSort(stocks, n);
63
64     cout << "Stock list after updates and sorting by change:\n";
65     printStocks(stocks, n);
66
67     return 0;
68 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

OUTPUT

[Running] cd "c:\Users\soumy\OneDrive\Desktop\DSA\" && g++ stock_ticker.cpp -o stock_ticker && "c:\Users\soumy\OneDrive\Desktop\DSA\stock_ticker"

Initial stock list:

Symbol	Price	Change

AAPL	178.23	+0
GOOGL	134.56	+0
TSLA	623.5	+0
NFLX	290.1	+0
AMZN	3450	+0

Stock list after updates and sorting by change:

Symbol	Price	Change

AMZN	3460	+10
GOOGL	135	+0.44
AAPL	178.23	+0
NFLX	290.1	+0
TSLA	620	-3.5

No forwarded ports. Forward a port to access your locally running services over the internet.

Forward a Port



> OUTLINE
> TIMELINE



0 3 0

3 Breaking points

Ln 64, Col 65 Spaces: 4 UTF-8 CRLF () C++ Win32

ENG 23:49